

# Lab: Derivation Exercise — Generative Models and Variational Inference

## Objective

Derive the variational inference update equations for a concrete generative model, connecting the abstract mathematics to explicit computations.

## Part 1: Hidden Markov Model Specification

**Goal:** Write out a complete generative model.

Consider a Hidden Markov Model with:

- $K = 3$  hidden states ( $s \in \{1, 2, 3\}$ )
- Observation model:  $p(o_t | s_t = k) = \mathcal{N}(o_t; \mu_k, \sigma_k^2)$  with known parameters
- Transition model:  $p(s_t | s_{t-1})$  given by a  $3 \times 3$  matrix  $\mathbf{A}$  with rows summing to 1
- Prior:  $p(s_1) = \text{categorical}$  with probabilities  $[0.5, 0.3, 0.2]$

**Task:** Write the complete joint distribution  $p(o_{1:T}, s_{1:T})$  for  $T = 3$  timesteps. Expand the product notation fully.

*Write your response here*

## Part 2: Variational Free Energy for the HMM

**Goal:** Derive the free energy for this model.

Let  $q(s_{1:T}) = \prod_{t=1}^T q(s_t)$  be the mean-field recognition density, where each  $q(s_t)$  is a categorical distribution with parameters (probabilities)  $r_t = [r_{t1}, r_{t2}, r_{t3}]$ .

**Task:** Write the variational free energy:

$$F = E_q[\ln q(s_{1:T})] - E_q[\ln p(o_{1:T}, s_{1:T})]$$

Expand each term. The first term is the negative entropy of  $q$ . The second term involves the log-likelihood and log-transition probabilities.

*Write your response here*

## Part 3: Optimal Update Equations

**Goal:** Derive the fixed-point equations for the sufficient statistics.

Taking the derivative  $\partial F / \partial r_{ij} = 0$  (with the constraint  $\sum_j r_{ij} = 1$ ), derive the optimal  $q^*(s_i)$ :

$$\ln q^*(s_i = k) \propto \ln p(o_i | s_i = k) + E_{-}\{q(s_{i-1})\} [\ln p(s_i = k | s_{i-1})] + E_{-}\{q(s_{i+1})\} [\ln p(s_{i+1} | s_i = k)]$$

Show all steps. Explain why the update depends on messages from both the past ( $s_{i-1}$ ) and the future ( $s_{i+1}$ ).

*Write your response here*

## Part 4: Gaussian State-Space Model

**Goal:** Repeat the derivation for a continuous model.

Consider a linear Gaussian state-space model:

- $p(s_t | s_{1:t-1}) = N(s_t; A \cdot s_{t-1}, Q)$  — state dynamics
- $p(o_t | s_t) = N(o_t; C \cdot s_t, R)$  — observation model
- $p(s_1) = N(s_1; \mu_0, \Sigma_0)$  — prior

With recognition density  $q(s_t) = N(s_t; m_t, V_t)$ .

**Task:** Derive the update equations for the sufficient statistics  $m_t$  (mean) and  $V_t$  (covariance) by minimizing F. Show that these correspond to Kalman filter/smoothing equations.

*Write your response here*

## Part 5: Synthesis

In 200 words, explain the common mathematical structure across Parts 2-4: in both discrete and continuous cases, the optimal recognition density arises from minimizing free energy, and the update equations combine likelihood (data-driven) terms with transition (model-driven) terms.

*Write your response here*

## Lab Summary

| Part | Skill Developed        | Mathematical Result                     |
|------|------------------------|-----------------------------------------|
| 1    | Model specification    | Writing the HMM joint distribution      |
| 2    | Free energy derivation | Expanding $F$ for mean-field HMM        |
| 3    | Variational calculus   | Deriving optimal update equations       |
| 4    | Continuous extension   | Gaussian state-space / Kalman filtering |
| 5    | Conceptual synthesis   | Unifying discrete and continuous cases  |